Norwegian School of Economics
Bergen, November 2023

BAN402 Decision Modelling in Business

Project 3

Candidate numbers: 39 **and** 115

Contents

PART A

In this task, we choose to slightly alter the `.dat` file, changing it from using `1993K1-2022K4` to using `1-120`. This makes it easier to incorporate the time aspect in the model. We us the set T, $\forall t \in T$ with values of t from 1 to 120.

In your report, include a screenshot of your .mod file

```
set T;
param L_init;
param Trend_init;
param PIFED {T};
var L{T} >= 0;
var Trend{T};
var F{T} >= 0:;
var gamma >=0;
var beta >=0;
minimize MAPE:
  (100 / card(T)) * sum {t in T: t>1 and t<=120}
    abs((PIFED[t] - F[t]) / PIFED[t]);
s.t.
LevelEquation {t in T: t>1 and t<=120}:
    L[t] =  gamma * PIFED[t] + (1 - gamma) * (L[t-1] + Trend[t-1]);
TrendEquation {t in T: t>1 and t<=120}:
    Trend[t] = beta * (L[t] - L[t-1]) + (1 - beta) * Trend[t-1];
ForecastEquation {t in T: t>1 and t<120}:
    F[t+1] = L[t] + Trend[t];
gamma1: gamma >= 0.01;
gamma2: gamma <= 0.99;
beta1: beta >= 0.01;
beta2: beta <= 0.99;
L1:     L[1] == L_init;
Trend1: Trend[1] == Trend_init;
```

Task a: What is the optimal value of MAPE?

The optimal value for MAPE is

$$\text{MAPE} = \frac{100\%}{120} \sum_{t=1}^{120} \left| \frac{PIFED_t - F_t}{PIFED_t} \right| = 2.11889 \tag{1}$$

Task b: What are the optimal values of γ and β?

The optimal values are $\beta = 0.0254784$ and $\gamma = 0.721195$.

Task c: Among the 120 quarters, what is the maximum APE and how much is it?

The maximum APE is in quarter 64 with a value $APE = 10.2785$.

Task d: Among the 120 quarters, what is the minimum APE and how much is it?

The minimum APE is in quarter 62 with a value $APE = 3.3356e - 12$.

Task e: What would be your forecast for the PIFED of the first quarter of 2023?

We have the following equations from the project description:

$$F_{t+1} = L_t + Trend_t \tag{2}$$

$$L_{t+1} = \gamma \cdot PIFED_{t+1} + (1 - \gamma) \cdot (L_t + Trend_t) \tag{3}$$

$$Trend_{t+1} = \beta \cdot (L_{t+1} - L_t) + (1 - \beta) \cdot Trend_t \tag{4}$$

We know the following values for $t = 120$:

$$\beta = 0.0254784 \qquad \gamma = 0.721195 \qquad F_{120} = 145.092 \tag{5}$$

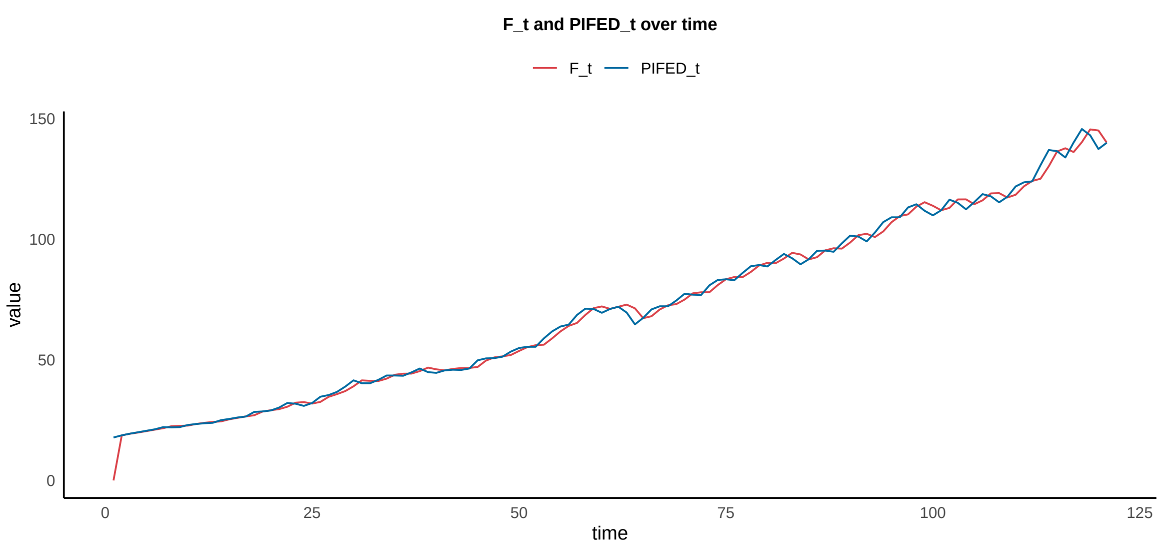
$$L_{120} = 139.544 \qquad Trend_{120} = 1.17048 \tag{6}$$

Finding F_t with $t = 121$.

$$F_{121} = L_{120} + Trend_{120} = 139.544 + 1.17048 = 140.71 \tag{7}$$

We find $F_{121} = 140.71$, which is 0.71 off the actual *Statistics Norway* value at 140.

Task f: Plot forecast and PIFED values from 1993 to first quarter of 2023.



PART B

Task 1: No district can have more than six neither less than six transplant centers than the average number of transplant centers across districts.

Given that $H = \sum_{i \in I} h_i$ is the total number of transplant centers across all DSAs, and N is the number of districts, the average number of transplant centers per district is $\bar{h} = \frac{H}{N}$. The constraints for each district k can be formulated as:

$$\sum_{i \in I} h_i W_{ik} \leq \bar{h} + 6, \quad \forall k \in K \tag{1}$$

$$\sum_{i \in I} h_i W_{ik} \geq \bar{h} - 6, \quad \forall k \in K \tag{2}$$

Where W_{ik} is a binary decision variable that is 1 if DSA i is in the district with center at DSA k, and 0 otherwise.

Task 2: The health authorities require that your solution can contain at most two strong districts.

Let the set $\mathcal{J} = \{1, 2, \ldots, 10\}$ denote the indices of the DSAs in the Top 10 list for liver recoveries. We introduce binary variables S_k for each district k to indicate whether a district is "strong" (having three or more of the top 10 DSAs). The model constraints to limit the number of strong districts to at most two are formulated as follows:

$$\sum_{i \in \mathcal{J}} W_{ik} \leq 2 + M * S_k, \quad \forall k \in K \tag{3}$$

$$\sum_{i \in \mathcal{J}} W_{ik} \geq 3S_k, \quad \forall k \in K \tag{4}$$

The above equations where W_{ik} is 1 if DSA i is assigned to the district centered at DSA k, and 0 otherwise. M is a large number, more than total possible DSAs (big M). If a district k is a strong district, S_k must be 1.

$$\sum_{k \in K} S_k \leq 2 \tag{5}$$

The above equation restricts the number of strong districts to no more than two.

$$S_k \in \{0, 1\}, \quad \forall k \in K \tag{6}$$

The above equation ensures that S_k variables are binary, indicating the presence or absence of a strong district condition.

PART C

Having candidate numbers 39 and 115, we use $5/2 = 2.5 \approx 3$ in our project.

Task 1

Draw figures based of the supply and demand curves for one of these periods.

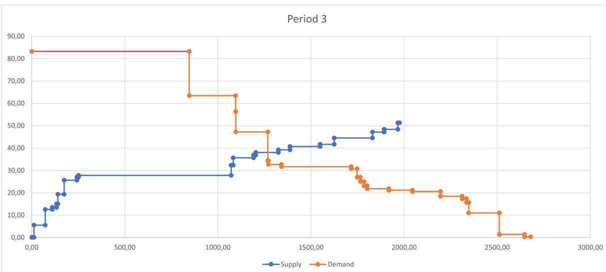


Figure 1: Step function curve

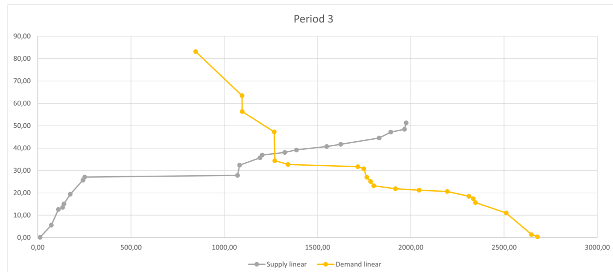


Figure 2: Linearized curve

Based on your figure, which price (approximately) would you propose as the system price for this period and which bids should be accepted?

Based on the stepwise figure, we would propose a price of approximately somewhere between €48 and €34 with a total volume of 1250 for period 3. Buyer bids around 48€ and seller bids around 34€ should therefore be accepted. Based on the linear figure, we would propose a price of approximately €38 with a total volume of 1250 for period 3. For the linear figure, bids around the price €38 should be accepted.

Advantages/disadvantages with the step function and linearized curves.

Step function curves in this represents the accurate demand and supply for price and quantities. This is a more accurate depiction, but may make it harder to decide an accurate equilibrium price. Linearized curves are less accurate in representing the specific demand and supply for price and quantities. On the upside, linearized curves are better at deciding an accurate equilibrium price. With the linearized curves, there is always some error, over and underrepresenting the prices for demand and supply respectively.

Task 2a

For the tables in this task Q_{bid} is the quantity we offed, P_{bid}€ is our bid price, Q_{acc} is the accepted quantity for the last bid, P^*€ is the equilibirum price, C€ is the cost, R€ is the revenue, and Π€ is the profit. **The profit we obtain is:** 62700.

Reasoning behind our strategy:

The first strategy we use is to try selling the maximum quantity possible. We do this by progressively lowering the price, while keeping quantity fixed at $Q_{bid} = 1000$. We use this as a benchmark, and thereafter lower our quantity by a hundred looking to increase our profits.

Hour	Q_{bid}	P_{bid}€	Q_{acc}	P^*€	C€	R€	Π€
1	1000	7	1000	13.28	10	13280	3280
2	1000	9	1000	18.31	10	18310	8310
3	1000	27	1000	27.54	10	27540	17540
4	1000	21	1000	21.23	10	21230	11230
5	1000	32	1000	32.34	10	32340	22340
$\sum$						**112700**	**62700**

In this above strategy we find that by lowering the prices quite drastically, we are able to sell the full quantity of $Q = 1000$. Calculating the profit we earn Π€=62700. Next, we try lowering the quantity to 900, and use the same strategy as in selling the maximum quantity. This strategy yields a profit of Π€=59625, being worse than before.

Hour	Q_{bid}	P_{bid}€	Q_{acc}	P^*€	C€	R€	Π€
1	900	14	900	14.74	10	13266	4266
2	900	13	900	19.51	10	17559	8559
3	900	27	900	27.62	10	24858	15858
4	900	21	900	21.58	10	19422	10422
5	900	31	900	32.80	10	29520	20520
$\sum$						**104625**	**59625**

Task 2b

In task 2a, we previously concluded that uniformly decreasing quantities across all time frames would not be advantageous compared to the initial scenario. With the flexibility to adjust bid quantities for individual periods now available, we can develop a methodical approach to identify which periods, if any, might gain from a reduction in quantity. Specifically, this method will incrementally aid us in pinpointing the extent of the reduction needed for each period. We will incorporate ask price adjustments into our strategy to work alongside this plan. As the quantity lessens, the requisite increase in the linear equilibrium price to maintain the same level of profits as initially expected becomes greater. Therefore, more substantial price changes at the beginning of the process are more likely to be financially beneficial. Consequently, we stop the iteration once we reach a quantity reduction of 400. The step-by-step progression of this approach is documented in the appendix 1.

Period 1: In the initial phase of our analysis, we note a significant adjustment in the linearized equilibrium price precipitated by a reduction of 200 units in quantity. Subsequent iterations, adjusting the quantity in increments and decrements of ten, reveal a stable equilibrium price at a quantity of 790, with a noticeable price decline at 810. Consequently, we establish that the ideal quantity for this period is 800 units. This increased price is adequately significant to offset the loss incurred by the amount reduced. In instances where the entire bid finds acceptance, fine-tuning the ask price becomes necessary. By raising the asking price, we can effectively moderate the accepted quantity while maintaining a stable equilibrium price.

Period 2: Upon examining period two, we discover that reducing the quantity by 300, alongside price tuning, yields better profit than our original bid. However, it's worth noting that even a substantial increase in price doesn't guarantee profitability. This prompted us to conduct a local search to determine whether an increase in quantity could still capitalize on the significant price rise without sacrificing profits. Through iterative adjustments by increments of ten, we pinpointed the optimal amount to be 710.

Periods 3-5: In the case of periods three, four, and five, our analysis indicates that reducing the quantity does not enhance the outcome; therefore, we elect to retain the original bid quantity. This decision is bolstered by our observations from iterative testing, where decreasing the amount in steps of ten from the original does not lead to significant price variation that can beat the initial profit level.

Task 3: Decide the best block bid in order to maximize your profit.

For this task, we need to choose blockbids for either periods 1-4 or 2-5. Looking at the market before our blockbid, we find that periods 1-4 has higher prices and larger quantities compared to periods 2-5.

Hour	Price	Volume	PS	PD	s	d
1	36.44	968.00	34.90	53.30	34.00	146.00
2	32.11	1298.00	25.80	34.10	66.00	22.00
3	33.24	1271.00	27.80	34.40	811.00	3.00
4	33.32	1259.00	33.10	36.50	64.00	1.00
5	21.03	857.00	18.80	23.70	203.00	15.00

Reasoning behind our strategy: We may try to get our maximum quantity of 1000 accepted. One strategy as such would be to start with the average price of periods 1-4, doing so we find that the average equilibrium price is

$$(36.44 + 32.11 + 33.24 + 33.32)/4 = 33.78 \tag{1}$$

Further, the average last accepted supply price is:

$$(34.90 + 25.80 + 27.80 + 33.10)/4 = 30.4 \tag{2}$$

Using this intuition, we find that rounding down the equilibrium price to 33.7 accepts our blockbid.

Price	Volume	id	begin	end	order
19.40	20.00	2	2	4	1
31.40	175.00	3	1	5	3
25.50	14.00	6	2	4	2
33.70	1000.00	11	1	4	4

The profit we obtain is:

For this benchmark, the profit we obtain is Π€= 46660. In the appendix we try to increase this profit by undercutting the other blockbids of respectively 31.40, 25.50, and 19.40. Doing so, we find a maximum profit of Π€= 58410 when bidding $Q = 1000$ at P_{bid}€=25.40 for period 1-4. This is shown in the appendix in 2.

PART D

Sets:

K : Set of counties
J : Set of municipalities
$\bar{J}_k \subseteq J$: Subset of municipalities belonging to county $k, \forall k \in K$
I : Set of companies
B : Set of bids, where each bid includes a price and one or more municipalities

Parameters

P_b : Price of each bid $b \in B$
D_i : Discount offered by company $i \in I$
T_i : Threshold parameter for discount by company $i \in I$
maxN : Maximum number of companies allowed
a_{ib} : Binary parameter, 1 if bid b is by company i, 0 otherwise
m_{jb} : Binary parameter, 1 if municipality j is in bid b, 0 otherwise
$BigM1$: The value of Big M is equal to the cardinality of the bid set
$BigM2$: The value of Big M is equal to the cardinality of the county set

Decision Variables

$x_b \in \{0,1\}, \forall b \in B$	: 1 if bid b is accepted, 0 otherwise
$y_i \in \{0,1\}, \forall i \in I$	: 1 if the discount for company i is applied, 0 otherwise
$z_{ik} \in \{0,1\}, \forall i \in I, k \in K$	: 1 if company i has an accepted bid in county k, 0 otherwise
$u_i \in \{0,1\}, \forall i \in I$	: 1 if company i has one accepted bid or more, 0 otherwise

Objective: minimize $\text{Total\_Cost} = \sum_{b\in B} P_b x_b - \sum_{i\in I} D_i y_i$

Constraints

$$\begin{aligned}
&\text{Municipality_Coverage}_{j\in J}: && \textstyle\sum_{b\in B} m_{j,b} \cdot x_b \geq 1;\\
&\text{lb-Z}_{i\in I,k\in K}: && z_{i,k} \leq \textstyle\sum_{b\in B} a_{i,b} \cdot \left(\sum_{j\in \bar{J}_k} m_{j,b}\right) \cdot x_b;\\
&\text{ub-Z}_{i\in I,k\in K}: && z_{i,k} \geq \frac{1}{\text{BigM1}} \cdot \textstyle\sum_{b\in B} a_{i,b} \cdot \left(\sum_{j\in \bar{J}_k} m_{j,b}\right);\\
&\text{County_Service}_{k\in K}: && \textstyle\sum_{i\in I} z_{i,k} \geq 2;\\
&\text{LinkUtoZ}_{i\in I}: && u_i \geq \frac{1}{BigM2} \cdot \textstyle\sum_{k\in K} z_{i,k};\\
&\text{SetU}_{i\in I}: && \textstyle\sum_{k\in K} z_{i,k} \geq u_i;\\
&\text{Company_Limit}: && \textstyle\sum_{i\in I} u_i \leq \text{maxN};\\
&\text{Discount_Application}_{i\in I}: && \textstyle\sum_{b\in B} a_{i,b} \cdot P_b \cdot x_b \cdot u_i \geq T_i \cdot y_i;
\end{aligned}$$

We have conducted a model validation exercise by implementing it in AMPL and running it on a sample dataset, even though it was not mandatory. The files related to this exercise are stored in the "D" folder.

Block Bids and Economies of Scale: In our model, the binary parameter $a_{i,b}$ plays a crucial role, enabling companies to submit bids for various services. This parameter is set to 1 if a particular bid b is offered by company i, and 0 otherwise. Concurrently, the binary parameter $m_{j,b}$ is utilized to signify whether a bid b includes a specific municipality j. The use of $m_{j,b}$ allows for the inclusion of multiple municipalities within a single bid, facilitating block bids that can leverage economies of scale. For instance, if a company submits a bid that encompasses several municipalities, it might be able to lower its operational costs due to synergies across these areas, which is advantageous for both the company and the contracting authority. Thus, when a bid b (indicated by $x_b = 1$) is accepted, it ensures service provision to all municipalities included in that bid. Our model is designed to encourage the submission of such inclusive bids, promoting cost efficiencies similar to those observed in larger, more aggregated market setups.

Appendix

Appendix 1: Part C Task 2b

Table 1: Iterations of bids for periods 1-5. Optimal strategy marked in green.

Iteration	Period	Q_{bid}€	Q_{acc}€	P_{bid}€	P^*€	C€	R€	Π€
Original	1	1000	1000	7	13.28	10	13,280	3,280
1.1	1	900	900	14	14.28	10	12,852	3,852
1.2	1	810	810	16	16.39	10	13,276	5,176
1.3	1	800	800	16	16.5	10	13,200	5,200
1.4	1	790	790	16	16.5	10	13,035	5,135
1.5	1	700	700	4	16.67	10	11,669	4,669
1.6	1	600	600	14	17.77	10	10,662	4,662
Original	2	1000	1000	9	18.31	10	18,310	8,310
2.1	2	900	900	13	19.51	10	17,559	8,559
2.2	2	800	800	11	20.78	10	16,624	8,624
2.3	2	720	720	22	22.87	10	16,466	9,266
2.4	2	710	710	11	24.14	10	17,139	10,039
2.5	2	700	700	24	24.27	10	16,989	9,989
2.6	2	690	690	24	24.39	10	16,829	9,929
2.7	2	600	600	25	25.65	10	15,390	9,390
Original	3	1000	1000	27	27.54	10	27,540	17,540
3.1	3	990	990	27	27.54	10	27,265	17,365
3.2	3	900	900	27	27.62	10	24,858	15,858
3.3	3	800	800	27	27.71	10	22,168	14,168
3.4	3	700	700	27	27.79	10	19,453	12,453
3.5	3	600	600	27	31.79	10	19,074	13,074
Original	4	1000	1000	21	21.23	10	21,230	11,230
4.1	4	990	990	21	21.25	10	21,038	11,138
4.2	4	900	900	22	22.03	10	19,827	10,827
4.3	4	800	800	22	22.24	10	17,792	9,792
4.4	4	700	700	22	23.7	10	16,590	9,590
4.5	4	600	600	22	24.73	10	14,838	8,838
Original	5	1000	1000	32	32.24	10	32,240	22,240
5.1	5	990	990	32	32.3	10	31,977	22,077
5.2	5	900	900	32	32.84	10	29,556	20,556
5.3	5	800	800	36	36.02	10	28,816	20,816
5.4	5	700	700	36	36.09	10	25,263	18,263
5.5	5	600	600	36	36.17	10	21,702	15,702

Appendix 2: Part C Task 3

Table 2: Sensitivity analysis of blockbids for periods 1-4. Optimal strategy marked in green.

Q	P_{bid}€	P_1€	P_2€	P_3€	P_4€	C€	R€	Π€	Strategy
1000	33,7	28,96	11,65	27,36	18,69	10	86660	46660	max Q
1000	31,4	28,96	11,65	27,36	18,69	10	86660	46660	max Q similar to 31.4
1000	31,3	32,03	17,81	27,51	20,66	10	98010	58010	max Q undercut 31.4
1000	25,5	32,03	17,81	27,51	20,66	10	98010	58010	max Q similar to 25.5
1000	25,4	32,03	18,06	27,52	20,8	10	98410	58410	max Q undercut 25.5
1000	19,4	32,03	18,06	27,52	20,8	10	98410	58410	max Q similar to 19.4
1000	19,3	32,03	18,06	27,52	20,8	10	98410	58410	max Q undercut 19.4
1000	1	32,03	18,06	27,52	20,8	10	98410	58410	max Q go to limit 1
990	25,4	32,11	18,19	27,53	20,89	10	97732,8	58132,8	max Q-10
980	25,4	32,18	18,31	27,54	20,99	10	97039,6	57839,6	max Q-20
970	25,4	32,35	18,43	27,54	21,09	10	96427,7	57627,7	max Q-30
960	25,4	32,33	18,55	27,55	21,18	10	95625,6	57225,6	max Q-40
950	25,4	32,14	18,67	27,56	21,28	10	94667,5	56667,5	max Q-50
940	25,4	32,49	18,79	27,57	21,33	10	94169,2	56569,2	max Q-60
930	25,4	32,56	18,91	27,58	21,37	10	93390,6	56190,6	max Q-70
920	25,4	32,64	19,03	27,59	21,42	10	92625,6	55825,6	max Q-80
910	25,4	32,72	19,15	27,6	21,46	10	91846,3	55446,3	max Q-90
900	25,4	32,8	19,27	27,6	21,5	10	91053	55053	max Q-100
890	25,4	32,87	19,39	27,61	21,54	10	90254,9	54654,9	max Q-110
880	25,4	32,95	19,51	27,62	21,58	10	89460,8	54260,8	max Q-120
870	25,4	33,15	19,62	27,63	21,63	10	88766,1	53966,1	max Q-130
860	25,4	33,56	19,74	27,64	21,67	10	88244,6	53844,6	max Q-140
850	25,4	33,96	19,86	27,65	21,71	10	87703	53703	max Q-150